

國立高雄大學 114 學年度轉學招生考試試題(轉二年級)

科目：微積分
考試時間：80 分鐘

系所：
電機工程學系(無組別)
本科原始成績：100 分

是否使用計算機：是

1. Find $\lim_{x \rightarrow 0} \left(\frac{1 - \sec x}{x^2} \right)$. (10%)
2. Find $\frac{d}{dx} [\sin(7x^2 + x + 3)^3]$. (10%)
3. Please describe the process of finding extrema on a closed interval $[a, b]$. (10%)
4. Find $\int (x + 1) \sqrt{(x^2 + 2x + 5)} dx$. (10%)
5. Find $\int_{-\frac{\pi}{7}}^{\frac{\pi}{7}} [(\sin x)^7 x^2 + (\sin x)^{15} x^8] dx$. (10%)
6. Find $\int \frac{\cot x}{\ln(\sin x)} dx$. (10%)
7. Find $\int \frac{3x^2}{\sqrt{9-x^6}} dx$. (10%)
8. $z = x^2 + y$. Find the total differential of z . (10%)
9. Find the gradient of $f(x, y) = xy + y$. (10%)
10. $\vec{F}(x, y, z) = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$. Find $\nabla \cdot \vec{F}(x, y, z)$. (10%)

科目：物理

系所：電機工程學系(無組別)

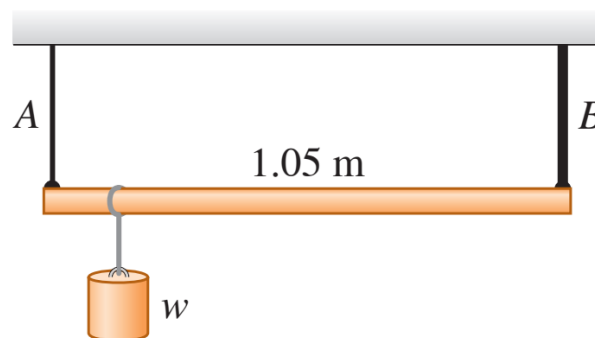
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考試時間：80 分鐘

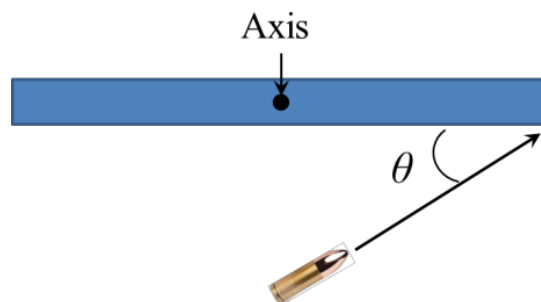
本科原始成績：100 分

*Gravitational acceleration, $g = 9.8 \text{ m/s}^2$

1. (7% + 8%) In the figure, A 1.05-m-long rod of negligible weight is supported at its ends by wires A and B of equal length. The cross-sectional area of A is 2.0 mm^2 and that of B is 4.0 mm^2 . Young's modulus for wire A is $1.8 \times 10^{11} \text{ Pa}$; that for B is $1.2 \times 10^{11} \text{ Pa}$. At what point along the rod should a weight w be suspended to produce (a) equal stresses in A and B and (b) equal strains in A and B?



2. (10%) A uniform thin rod with a length = 1.0 m and mass = 5.0 kg can rotate in a horizontal plane about a vertical axis through its center. The rod is at rest when a 2.5 g bullet traveling in the rotation plane is fired into one end of the rod. As viewed from above, the bullet's path makes an angle $\theta = 60^\circ$ with the rod as shown. If the bullet lodges in the rod and the angular velocity of the rod is 12.0 rad/s immediately after the collision, what is the bullet's speed just before impact?

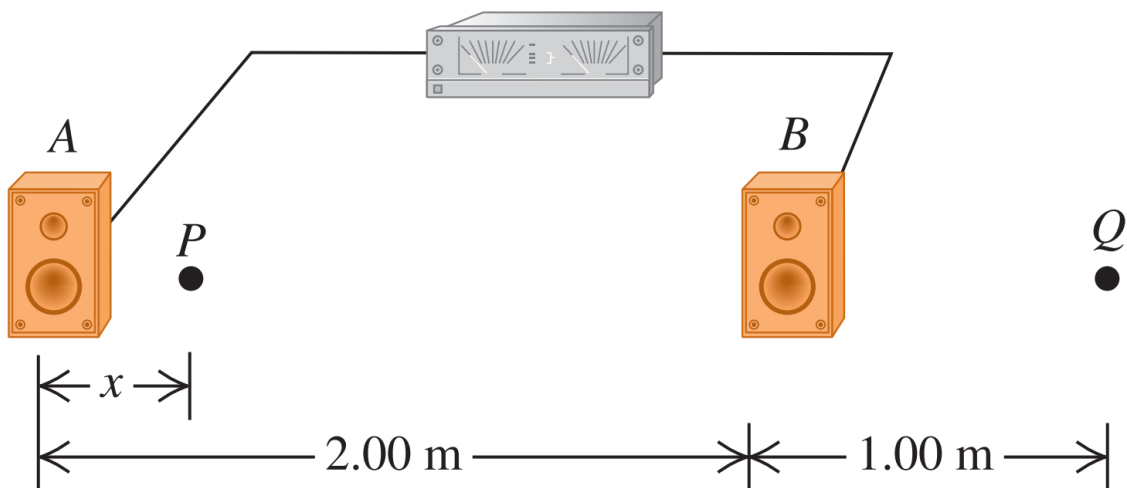


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3. (7% + 8%) Two loudspeakers, as shown in the figure, are driven by the same amplifier and emit sinusoidal waves in phase. Speaker B is 2.0 m to the right of speaker A.
- (a) If the frequency of the sound waves produced by the loudspeakers is 206 Hz. Consider a point P between the speakers and along the line connecting them, a distance x to the right of A. Both speakers emit sound waves that travel directly from the speaker to point P . For what values of x will destructive interference occur at P .
- (b) Consider point Q along the extension of the line connecting the speakers, 1.0 m to the right of speaker B. Both speakers emit sound waves that travel directly from the speaker to point Q . What is the lowest frequency for which constructive interference occurs at point Q .

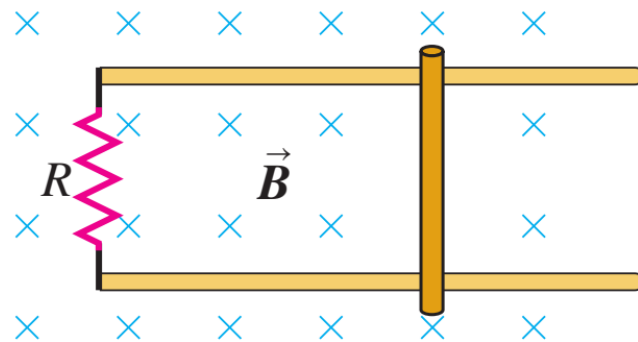


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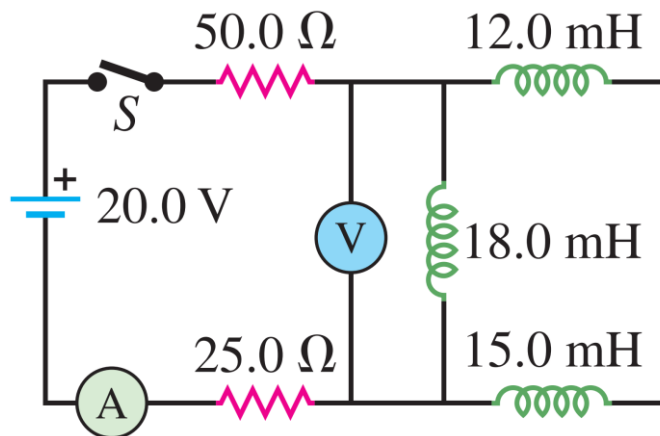
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4. (15%) A 0.25-m-long bar moves on parallel rails that are connected through a $6.0\ \Omega$ resistor, as shown in the Figure, so the apparatus makes a complete circuit. You can ignore the resistance of the bar and rails. The circuit is in a uniform magnetic field $B = 1.2\ \text{T}$ that is directed into the plane of the figure. At an instant when the induced current in the circuit is counterclockwise and equal to $1.75\ \text{A}$, what is the velocity of the bar (magnitude and direction)?



5. (7% + 8%) As shown in the figure, the switch S has been open for a long time and is suddenly closed. Neither the battery nor the inductors have any appreciable resistance. What do the ammeter and the voltmeter read (a) Just after S is closed; (b) After S has been closed for a very long time?

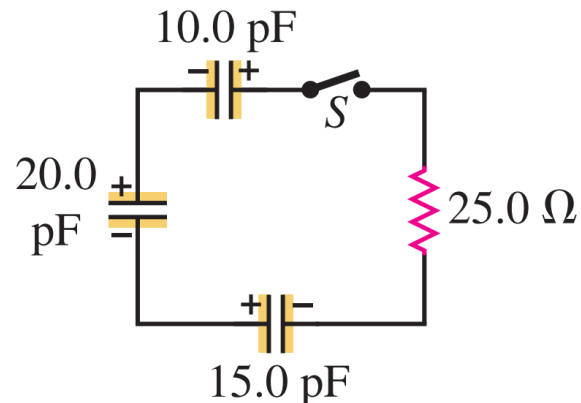


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6. (15%) In the circuit shown in the figure, each capacitor initially has a charge of magnitude 3.50 nC on its plates. After the switch S is closed, what will be the current in the circuit at the instant that the capacitors have lost 80.0% of their initial stored energy?



7. (7% + 8%) A particle with mass m and positive charge q starts from rest at the origin as shown in the figure. There is a uniform electric field $\mathbf{E}$ in the $+y$ -direction and a uniform magnetic field $\mathbf{B}$ directed out of the page. It is shown in more advanced books that the path is a *cycloid* whose radius of curvature at the top points is twice the y -coordinate at that level. (a) Prove that the speed at any point is equal to $\sqrt{12qEy/m}$. (b) Applying Newton's second law at the top point and taking as given that the radius of curvature here equals $2y$, prove that the speed at this point is $2E/B$.

